EEE2004: Digital Signal Processing

Tutorial 4

(February 19, 2026)

1. Determine the RoC of the z-transform of the following sequences without explicitly finding $X(z)$; in each case specify whether or not the unit circle is part of the RoC.
   (a) $x[n] = \left[\left(\frac{1}{2}\right)^n - \left(-\frac{4}{3}\right)^n\right] u[n-10]$
   (b) $x[n] = \left(-\frac{10}{9}\right)^n u[-n+4]$
   (c) $x[n] = \left(\frac{e}{\pi}\right)^n u[n+4]$
   (d) $x[n] = \left(\frac{e}{-\pi}\right)^n u[n] + (2+3j)^{n-2} u[-n-1]$
   (e) $x[n] = \sin(n)\, u[n-1]$
   (f) $x[n] = \dfrac{\sin(n)}{2^n} u[n-1]$

2. Consider the finite-duration sequence $h[n]$ defined in the range $0 \leq n \leq N-1$. Let $g[n] = h^*[N-1-n]$.
   (a) Express $G(z)$ in terms of $H(z)$.
   (b) Now let $g[n] = h[n]$, i.e., $h^*[N-1-n] = h[n]$. If z_0 is a zero of $H(z)$, can you find another zero related to z_0 ?
   (c) Suppose now $h[n]$ is real-valued and z_0 is a non-trivial zero that is not located on the unit circle. How many other zeros of $H(z)$ can you determine from the knowledge of z_0 ?

3. State which of the following are true/false. Justify your answer with proof if true, or with a counterexample if false:
   (a) For any sequence, whose z-transform exists and well-defined, its DTFT also exists and can be computed by evaluating the z-transform along the unit circle.
   (b) If two sequences $x_1[n]$ and $x_2[n]$ have well-defined z-transforms with RoCs R_1 and R_2 respectively, then any linear combination of the two sequences will have well-defined z-transform with RoC $R \subseteq R_1 \cap R_2$.
   (c) Suppose $X(z)$ is a well-defined z-transform of $x[n]$ with poles at $z = z_k$, where $k \in \{1, 2, \cdots, N\}$ and z_k's are non-zero and finite values. If $X(z)$ the number of possible RoCs is exactly two, then $z_1 = z_2 = \cdots = z_N$.

4. A causal LTI discrete-time system is described by the input/output equation $y[n] = 0.2\, y[n-1] + 0.08\, y[n-2] + 2\, x[n]$. Find the system transfer function $H(z)$.

5. Consider the transform pair $x[n] \overset{Z}{\longleftrightarrow} X(z)$ with RoC $|z| > r > 1$. For what range of values $a \in \mathbb{C}$ is the sequence $y[n] = a^n\, x[n]$ absolutely summable? Justify your answer.

6. Let $x[n] = 0.5^n u[n]$ and $y[n] = x[n] * x[n]$, where '$*$' denotes convolution. Evaluate $\displaystyle\sum_{n=-\infty}^{\infty} y[n]$.

7. Given $X(z) = \dfrac{z^{-1}}{(1-z^{-1})^2}$, determine the transform of the following signals using z-transform properties and specify the RoC (a) $x[n-3]$, (b) $x[n] \cdot \delta[n-3]$, (c) $x[n] - x[n-1]$, (d) $x[n] \cdot (\delta[n] - \delta[n-1])$, (e) $5\,x[n-1] + 4\left(-\dfrac{1}{3}\right)^n u[n]$.

8. Using the differentiation propery, find the sequences whose z-transforms are:
   (a) $X_1(z) = \log(1 - az^{-1})$ for $|z| > |a|$
   (b) $X_2(z) = \log\left(\dfrac{a}{a - z^{-1}}\right)$ for $|z| > \dfrac{1}{|a|}$
   (c) $X_3(z) = \log\left(1 - a^{-1}z\right)$ for $|z| < |a|$

9. Use z-transform properties to find the convolution result of $x[n]$ and $h[n]$, where $x[n] = \delta[n] - 2\,\delta[n-2]$ and $h[n] = 2\,\delta[n] - 2\,\delta[n-1] + 3\,\delta[n-2] + \delta[n-3]$.

10. Let $x[n] = (0.3)^n \exp(j0.25\pi n)\, u[n]$. The *autocorrelation* sequence $r_{xx}[n]$ corresponding to $x[n]$ is defined as $x[n] * x^*[-n]$. Find (a) $X(z)$, (b) the z-transform of $x^*[-n]$, (c) $R_{xx}(z)$ (by using properties), and (d) $r_{xx}[n]$ from $R_{xx}(z)$.